
Uncertainty-Aware Neural Surrogate for Parametric L-Bracket Stress Prediction

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Neural surrogates for finite-element analysis (FEA) promise large speedups for
2 design-space exploration but return predictions without telling the engineer when
3 to trust them. The missing ingredient is a calibrated reliability layer: a per-
4 prediction interval that flags when the surrogate should be trusted and when to
5 defer back to the full simulation. We build one for a parametric flat L-shaped corner
6 bracket (hereafter, the L-bracket) in 2D plane-stress linear elasticity, combining a
7 MeshGraphNets-style graph neural network (GNN) surrogate, a 5-member Deep
8 Ensemble, and conformalized quantile regression (CQR). On in-distribution (ID)
9 test data the surrogate achieves 1.81% per-node mean absolute percentage error
10 (MAPE), and the conformal layer meets its coverage guarantee at every tested nominal
11 level (80–95%). On a pre-registered out-of-distribution (OOD) set, accuracy
12 degrades and coverage drops below nominal, but the ensemble uncertainty signal
13 more than doubles — enabling a defer-to-FEA rule that detects 58% of the hardest
14 corner-OOD cases at a 5% false-alarm rate. The contribution is the evaluation
15 methodology — pre-registered OOD testing, per-condition extrapolation audit, and
16 a concrete trust-or-defer decision rule — not the individual components.

17 1 Introduction

18 FEA is the standard tool for stress prediction on structural parts, but its cost per evaluation rules it
19 out of tight-inner-loop design workflows [6, 8]. Neural surrogates reduce per-evaluation cost from
20 seconds or minutes to milliseconds on a consumer GPU; GNNs are a particularly natural fit because
21 they operate on the native FEA mesh [2, 6, 8]. A bare surrogate, however, returns a prediction for
22 every query in the design space, including queries for which it has no training evidence. The missing
23 ingredient is a calibrated reliability layer: a per-prediction interval whose coverage is documented
24 and whose width is informative, so that the engineer can trust the surrogate when the interval is tight
25 and defer to FEA when it is wide.

26 **Study part.** We study the L-bracket, a canonical thin shelf-bracket with three stress concentrators —
27 an inside fillet at the concave corner and two through-holes (Figure 1). The part is modelled in 2D
28 plane-stress linear elasticity under a static shelf load, with three free design parameters: the inside
29 fillet radius R , the horizontal-flange hole position p , and the flange width W .

30 **Contribution.** The contribution of this paper is the reliability layer, not the surrogate. We combine
31 three well-established ingredients — a MeshGraphNets-style GNN [6, 8], a Deep Ensemble for epis-
32 temic uncertainty [5], and split-conformal quantile regression [1, 3, 10] — into a single deployment-
33 calibrated surrogate for parametric L-bracket stress prediction. Gopakumar et al. [3] demonstrate
34 that cell-wise conformal prediction provides marginal coverage guarantees for neural PDE surrogates
35 across several architectures. We build on their framework in a stationary structural-mechanics setting
36 where the mesh varies between samples, and complement the coverage analysis with a pre-registered
37 out-of-distribution protocol, a per-condition extrapolation audit, and a deployment-oriented deferral



Figure 1: A flat L-shaped corner bracket in AISI 304 stainless steel, representative of the geometry studied in this work.

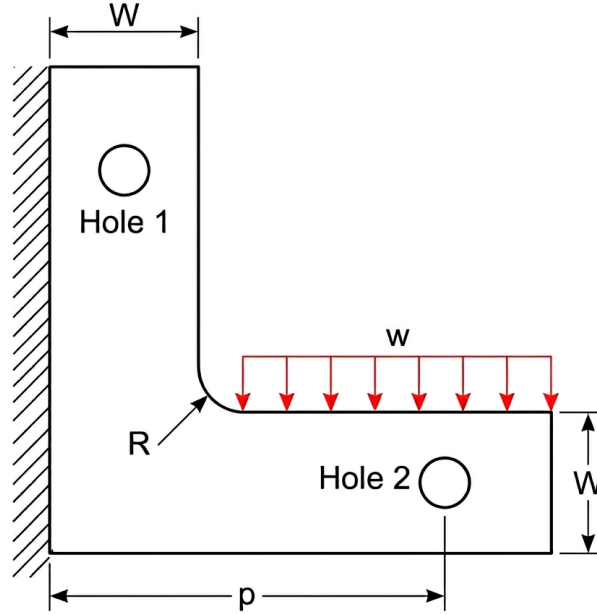


Figure 2: Geometry and boundary conditions of the parametric L-bracket. R : inside fillet radius; p : Hole-2 position from the clamped face; W : flange width (symmetric). The back face of the vertical flange ($x = 0$) is fully clamped; a uniform shelf traction w acts on the top face of the horizontal flange. (Lowercase w denotes the applied traction; uppercase W denotes flange width.)

rule with quantified operating characteristics. We verify the conformal coverage guarantee empirically at four nominal levels on a held-out ID test set, show that accuracy degrades and coverage drops on geometries outside the training box while ensemble disagreement more than doubles, and present an ensemble-size ablation that confirms the $M = 5$ choice.

2 Related Work

Graph neural networks operate directly on unstructured FEA meshes and predict solid-mechanics stress fields to within a few percent of the reference [2, 6, 8]. Uncertainty quantification for neural surrogates splits into Bayesian approximations and post-hoc calibration layers [9]; Deep Ensem-

Table 1: Parametric L-bracket problem setup.

Quantity	Value
Arm lengths $A = B$	80 mm
Thickness t (plane stress)	8 mm
Hole diameter D (both)	8 mm
Fillet radius R (swept)	[3.0, 10.0] mm
Hole-2 position p (swept)	[42.0, 72.0] mm
Flange width W (swept)	[14.0, 24.0] mm
Material	AISI 304 ($E = 193$ GPa, $\nu = 0.29$, $\sigma_y = 205$ MPa)
Distributed shelf traction w	0.8555 MPa (calibrated)
Clamped boundary	back face of vertical flange ($x = 0$)
FE scheme	FEniCSx, Lagrange order-2, $\sim 11,000$ degrees of freedom (DOF)
Sampling	Latin Hypercube sampling (LHS) [7], 5000 in-distribution + 250 OOD
Splits	4000 train / 500 val / 500 test / 250 OOD

bles [5] are the dominant practical choice for stress-field surrogates but the raw across-member standard deviation carries no formal coverage guarantee. Conformal prediction [1] converts any point or interval predictor into a prediction interval with finite-sample, distribution-free coverage under a single exchangeability assumption; CQR [10] sharpens the split-conformal interval by adapting its width to local heteroscedasticity. Gopakumar et al. [3] apply cell-wise conformal prediction to neural PDE surrogates (multilayer perceptron, U-Net, Fourier neural operator, Vision Transformer, GNN) on time-dependent spatio-temporal systems and study several nonconformity scores including CQR. We adopt their cell-wise pooling recipe in a mesh-native stationary structural-mechanics setting where the graph changes between samples, and extend the coverage analysis with a pre-registered OOD protocol, a per-condition extrapolation audit, and a deferral decision rule with quantified operating characteristics. The present work combines these ingredients: deep ensembles as the base predictor, CQR as the calibration wrapper, and cell-wise pooling for node-level marginal coverage on an FEA mesh.

3 Methods

3.1 FEA problem setup

The L-bracket is an equal-arm symmetric 2D plane-stress linear-elasticity problem with three variable geometric parameters; fixed outer dimensions, single AISI 304 stainless-steel material, and a static calibrated shelf traction are summarized in Table 1. The load magnitude $w = 0.8555$ MPa is calibrated in a preliminary unit-load run on the sample with the smallest fillet radius and narrowest flange width ($R = 3.0$ mm, $W = 14.0$ mm) — the geometry producing the highest stress concentration — so that the peak von Mises stress across the sweep stays at $\sim 50\%$ of σ_y , keeping the full training set in the elastic regime. Body forces, bolt-reaction contact at Hole 1, and the outside corner fillet are omitted: all three are expected to contribute negligibly to peak stress at this slenderness based on standard screening-level sizing practice. Body forces are three orders of magnitude below the applied traction; the outside corner lies on the compressive outer bend and does not concentrate stress; bolt-reaction contact at Hole 1 is subsumed by the clamped-face idealization. Mesh convergence and an analytical Euler–Bernoulli cross-check are reported in Appendix A.

3.2 Surrogate architecture and training

The surrogate is a MeshGraphNets-style encoder–processor–decoder [8] adapted to stationary stress prediction [6]. Each FEA sample is one graph with 13 per-node features (2D coordinate, 6-way boundary-class one-hot, the three design parameters (R, p, W) broadcast as global conditioning, and distances to the nearest fillet and hole-rim nodes) and 4 per-edge features: the relative displacement vector $(\Delta x, \Delta y)$, Euclidean length $\|\Delta\|$, and inverse length $1/\|\Delta\|$. The processor stacks $L = 5$ residual message-passing blocks at hidden width $H = 128$ with sum aggregation; a node MLP decodes per-node von Mises stress in MPa. Training minimizes Huber loss [4] ($\delta = 1$ MPa) with

Adam (lr 5×10^{-4} cosine-annealed to 10^{-5}), batch size 8, up to 60 epochs with patience 12 on validation loss. All training is performed on a single NVIDIA RTX 4060 Laptop GPU (8 GB).

3.3 Deep Ensembles and CQR

Following Lakshminarayanan et al. [5], we train a Deep Ensemble of $M = 5$ independently initialized members (differing only in random seed and dataloader shuffle); at inference the ensemble mean $\bar{\sigma}(\mathbf{x}_i)$ is the point estimate and the across-member standard deviation $s(\mathbf{x}_i)$ is the epistemic-uncertainty scalar.

We convert $(\bar{\sigma}, s)$ into base quantile predictors via a Gaussian ansatz, $\hat{q}_{\text{lo/hi}} = \bar{\sigma} \mp z_{1-\alpha/2} s$. For a linear-elastic problem with smooth parameter dependence, the ensemble-member predictions at each node are approximately normally distributed around the mean, so the Gaussian quantile rule is a reasonable starting point; the subsequent CQR calibration step corrects any residual miscalibration introduced by this assumption. We then apply split-conformal calibration (Algorithm 1 of Romano et al. [10]): on the 500-sample validation calibration set we compute the CQR nonconformity score $E_{ij} = \max(\hat{q}_{\text{lo}} - \sigma^*, \sigma^* - \hat{q}_{\text{hi}})$, pool scores across all nodes of all calibration samples into a single set (cell-wise pooling [3]: each node prediction is treated as an independent exchangeable unit), and take $\hat{Q}(\alpha) = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n}(\{E_{ij}\})$. The final interval is $[\hat{q}_{\text{lo}} - \hat{Q}, \hat{q}_{\text{hi}} + \hat{Q}]$, carrying the Theorem 1 finite-sample coverage guarantee under exchangeability. Our 500-sample validation set was previously used for a three-candidate hyperparameter selection on point accuracy, inducing a bounded optimistic bias on coverage: we acknowledge this reuse; the three-candidate selection limits the degree of adaptation.

3.4 Pre-registered OOD protocol

To make the extrapolation evaluation falsifiable, the OOD protocol was fixed in writing before any model was trained.¹ Each parameter is extended $\pm 20\%$ of its training span beyond the training bound (clipped at binding geometric constraints). 250 samples are allocated as 25 per single-parameter direction (6 directions, 150 total) plus 100 corner-OOD samples where ≥ 2 parameters are simultaneously outside the training box. All Phase-1 and Phase-2 metrics are re-computed on this set verbatim; no post-hoc exclusion is permitted.

3.5 Deferral decision rule

For deployment we threshold the per-sample mean ensemble standard deviation $\bar{s}(\mathbf{x})$: use the surrogate when $\bar{s} < T^*$, defer to FEA otherwise. The threshold is the 95th percentile of $\bar{s}$ on the ID test set, giving 5% ID false-alarm by construction.

4 Results

4.1 Surrogate accuracy and ensemble calibration

On the 500-sample ID test set, the ensemble-mean prediction attains **1.81%** per-node and **0.42%** peak-stress MAPE, with median absolute error 0.032 MPa and 99th percentile 0.32 MPa (full per-seed breakdown in Appendix B). The gap between the two MAPE figures is expected: per-node MAPE pools thousands of low-stress nodes where small absolute errors inflate percentage errors, whereas peak stress always occurs at the fillet in every training sample, so ensemble averaging at that consistently-located peak node reduces variance very efficiently. An ensemble-size ablation (Appendix G) confirms that $M = 5$ outperforms smaller ensembles on both accuracy and calibration quality. The central Phase-1 claim — that ensemble disagreement tracks actual error — is confirmed at both granularities: node-level Pearson 0.494 (Spearman 0.524) and sample-level Pearson **0.944** between ensemble std and absolute residual (Figure 3). Sample-level correlation is the number the Phase-2 CQR layer inherits.

¹The protocol was committed to the project version-control repository before any surrogate training began; the git commit timestamp serves as the pre-registration artifact. The protocol document and commit hash will be included in the public code release.

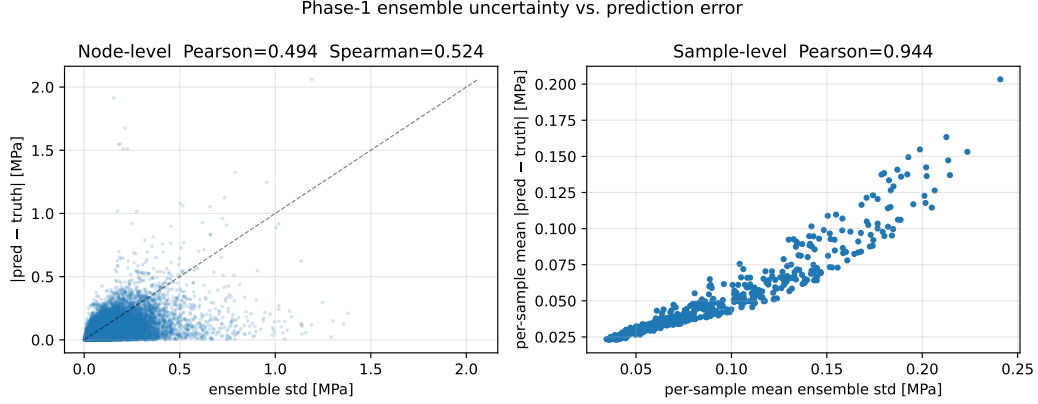


Figure 3: Ensemble uncertainty (x) vs. absolute error (y). *Left*: node-level over all test-set nodes. *Right*: per-sample scatter; each point is the mean uncertainty and mean error over one test sample. The strong sample-level trend is the empirical foundation for the Phase-2 CQR interval.

Table 2: Empirical coverage and mean interval width on the ID test set (pooled across $\sim 1.56 \times 10^6$ nodes), for the Gaussian ensemble interval $\bar{\sigma} \pm z_{1-\alpha/2}s$ and the CQR-calibrated interval; $\hat{Q}$ is the conformal correction learned on the 500-sample validation set. Full four-level sweep and conditional coverage in Appendix C.

Nominal [%]	Uncalibrated (Gaussian)		CQR-calibrated		$\hat{Q}$ [MPa]
	Coverage	Width [MPa]	Coverage	Width [MPa]	
80	86.41	0.25	82.72	0.24	-0.005
90	92.32	0.32	91.28	0.31	-0.003
95	95.19	0.38	95.53	0.38	0.001

4.2 CQR coverage on the held-out test set

Table 2 reports empirical coverage at three representative nominal levels for both the raw Gaussian ensemble interval and the CQR-calibrated interval, evaluated once on the held-out test set; the full four-level sweep (80/85/90/95) is in Appendix C.

At every tested level the CQR interval attains at or above its nominal coverage, consistent with Theorem 1 of Romano et al. [10] (Figure 4). The conformal correction $\hat{Q}$ is sub-MPa and signed: CQR tightens the over-wide small- α interval and widens the under-wide 95% interval. Conditional coverage sliced by parameter quartile stays within ± 3 pp of nominal across all twelve quartiles (Appendix C). Per-sample interval width correlates with per-sample absolute error at Pearson 0.944 — the deployment-relevant property (this correlation is inherited from the ensemble-std signal already plotted in Figure 3, since CQR adds a constant correction $\hat{Q}$ that does not affect the ranking of samples by width). Against a vanilla split-conformal baseline calibrated on the same validation pool (symmetric constant-width interval $\bar{\sigma} \pm Q_{\text{vanilla}}$, using $|\bar{\sigma} - \sigma^*|$ as the nonconformity score), CQR delivers coverage at nominal while vanilla under-covers (81.1% vs. 91.3% at $\alpha = 0.10$) despite using a narrower $2Q_{\text{vanilla}} = 0.167$ MPa interval — the adaptive CQR width absorbs the within-sample node correlation that the constant-width vanilla interval cannot. Full comparison in Appendix F. Having established that the CQR layer delivers calibrated intervals on in-distribution data, we now evaluate the full pipeline on geometries outside the training range.

4.3 Out-of-distribution evaluation

The CQR-calibrated ensemble, frozen after Phase 2, is evaluated verbatim on the 250 pre-registered OOD samples (Table 3). Three findings. (i) Accuracy degrades: per-node MAPE $1.7\times$, peak-stress MAPE $17\times$ (both ratios are full-OOD relative to ID); the disproportionate peak degradation reflects OOD geometries whose stress concentrators are sharper than any training sample. (ii) CQR coverage

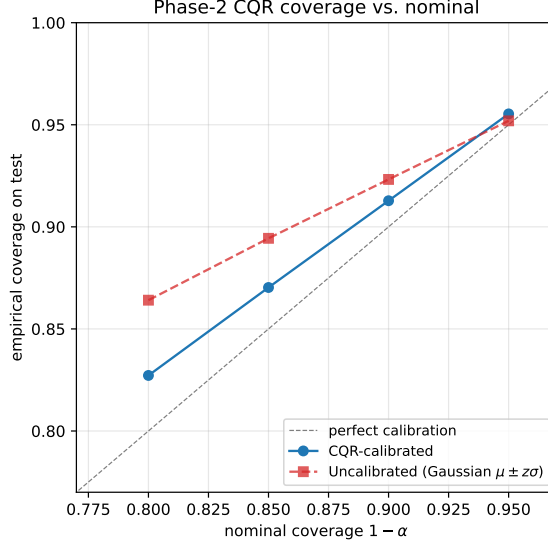


Figure 4: Empirical vs. nominal coverage. Diagonal is perfect calibration. CQR tracks the diagonal from above at every level, as required by the Theorem 1 guarantee.

Table 3: ID test vs. pre-registered OOD at nominal 90% coverage. ID: $N = 500, 1.56 \times 10^6$ nodes; OOD: $N = 250, 7.15 \times 10^5$ nodes. Per-condition OOD breakdown in Appendix D.

Metric	ID test	OOD	Change
Per-node MAPE [%]	1.81	3.10	$1.7\times$
Peak-stress MAPE [%]	0.42	7.26	$17\times$
CQR coverage at 90% [%]	91.28	83.70	-7.58 pp
CQR mean interval width [MPa]	0.311	0.654	$2.1\times$
Mean ensemble std [MPa]	0.096	0.208	$2.2\times$

148 drops below nominal (83.7% vs. 90%), the expected finite-sample realization of Theorem 1 losing its
 149 exchangeability premise. (iii) **The reliability layer scales up correctly**: mean CQR width roughly
 150 doubles and mean ensemble std more than doubles. The reliability layer produces a loud, measurable
 151 signal of degradation rather than failing silently.

152 The per-condition breakdown (Appendix D, Table 8) is informative: R -low and R -high extrapolations
 153 are handled essentially as well as ID ($\leq 1.3\%$ per-node MAPE, coverage within 2 pp, ensemble std
 154 at the ID baseline), while both p directions, W -low, and corner-OOD samples drive the aggregate
 155 degradation. Corner OOD shows 4.56% per-node MAPE ($2.5\times$ ID), coverage 81.5%, and $3.5\times$
 156 ensemble std.

157 4.4 Example predictions

158 Figure 5 shows best, median, and worst test samples ranked by per-sample mean absolute error.
 159 The fillet — the dominant stress concentrator — is recovered faithfully across the three quantiles;
 160 residual error concentrates on the fillet arc and hole rims, consistent with the literature-reported
 161 tail-compression pattern of GNN mesh surrogates [2, 6]. Critically, the per-node ensemble std (right
 162 column) tracks the error map: the GNN knows where it is less reliable.

163 4.5 Defer-to-FEA decision rule

164 At $T^* = 0.182$ MPa (ID 95th percentile of $\bar{s}$), the rule flags 5% of ID samples (false alarm, by
 165 construction) and detects 34% of the full OOD set and **58%** of the hardest corner-OOD condition.
 166 Per-condition detection is 0% on the R -direction perturbations (where the surrogate legitimately

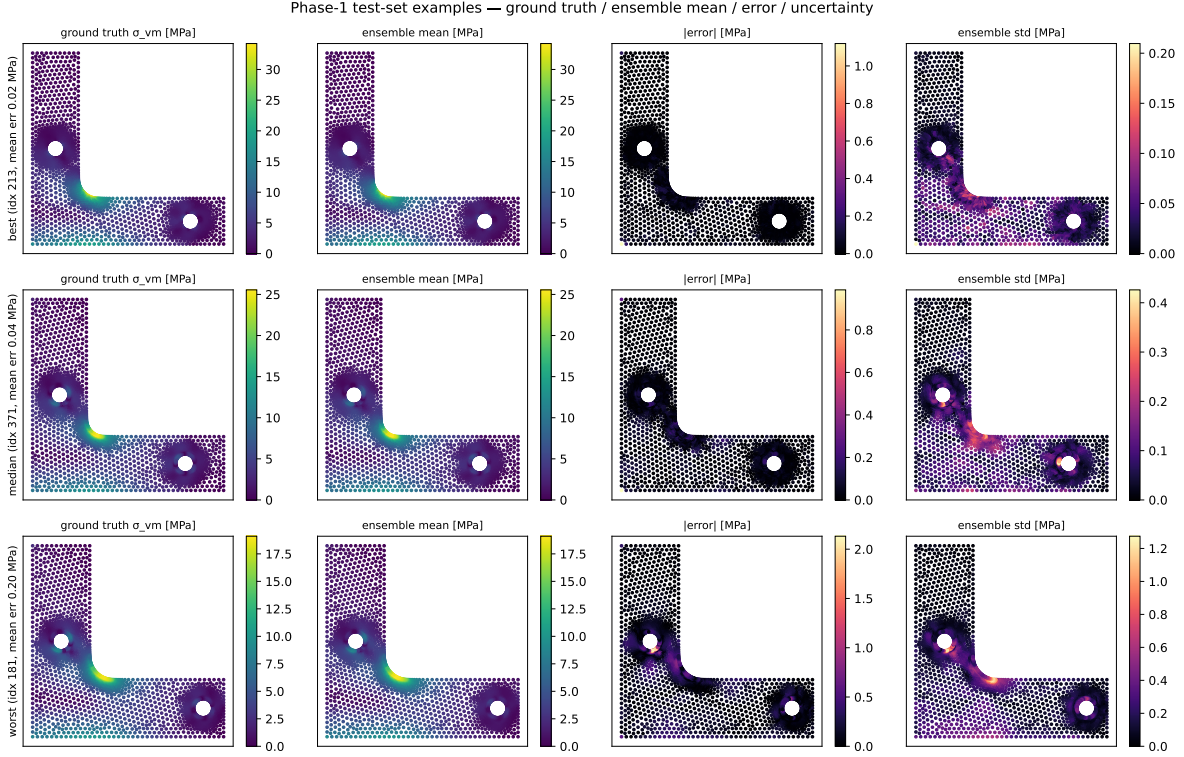


Figure 5: Test-set examples (best, median, worst by per-sample mean error). Columns: ground-truth von Mises stress; ensemble-mean prediction; absolute error; ensemble standard deviation. The uncertainty map tracks the error map.

extrapolates cleanly) and 4–44% on the remaining single-parameter conditions. The area under the receiver operating characteristic curve (AUROC) of $\bar{s}$ as an OOD detector is 0.673; some OOD conditions are in practice easier for the surrogate than the hardest ID quartile (Appendix E, Figure 13).

5 Discussion

What CQR adds. Ensemble disagreement alone is a scalar without a probabilistic interpretation: in principle, two ensembles with identical Pearson correlation to the truth can produce differently calibrated intervals. CQR converts that signal into a prediction interval with a distribution-free finite-sample coverage guarantee. For the L-bracket the raw Gaussian interval is already close to calibrated and the conformal correction remains sub-MPa — this is a friendly case where CQR’s contribution is modest in magnitude but categorical in interpretation.

Which extrapolation axes are safe. The per-condition breakdown (Appendix D) carries a concrete engineering message: on this geometry the surrogate extrapolates cleanly on R in both directions but not on p or W , and not at all on joint extrapolation. Any similar surrogate rollout should publish this per-axis extrapolation audit alongside the headline accuracy number.

Limitations. (i) Coverage is marginal, not conditional; covariate-shift conformal extensions [1] would formalize the OOD case. (ii) The calibration set doubles as the Phase-1 hyperparameter-selection set; adaptation is bounded but not zero. (iii) Cell-wise pooling treats mesh nodes within a graph as exchangeable, which they strictly are not. (iv) No aleatoric head: the FEA target is noise-free, so the reliability layer captures only epistemic uncertainty. (v) The physical model is idealized (2D plane stress, single material, linear elasticity, no self-weight); relaxing any of these requires re-calibration. (vi) OOD is $\pm 20\%$ parametric extrapolation; structural extrapolation (new topology) is out of scope.

Scalability. The conformal calibration and ensemble components add constant multiplicative overhead to the base surrogate and impose no architectural constraints. In our experiments, CQR calibration over 1.5 million pooled nonconformity scores completed in seconds, with FEA data generation remaining the computational bottleneck rather than the UQ layer. The framework thus transfers directly to larger meshes and additional ensemble members; empirical validation on higher-dimensional parametric families remains future work.

Future work. Asymmetric-arm brackets and higher-dimensional parametric families, multiple load cases, nonlinear material response, and 3D extension. A continuous covariate-shift-aware recalibration [1] would replace the binary deferral rule with a smoothly widened interval as the input drifts from the training distribution.

6 Conclusion

We built a deployment-calibrated neural surrogate for parametric L-bracket stress prediction by stacking a MeshGraphNets-style GNN, a 5-member Deep Ensemble, and a split-conformalized quantile-regression layer on a common FEA training set. The surrogate attains 1.81% per-node and 0.42% peak-stress MAPE; the conformal layer delivers the Romano 2019 Theorem 1 guarantee at every tested nominal level; the interval width is informative at Pearson 0.944 per sample. On a pre-registered OOD set the guarantee expires as predicted but the reliability layer’s own signal more than doubles, yielding a defer-to-FEA rule that detects 58% of corner-OOD samples at 5% ID false-alarm. The contribution is the combination and the honest evaluation under a pre-registered protocol: a reliability retrofit that is already deployable with today’s standard UQ machinery.

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Conference For AI Scientists 2026 - AI Involvement Checklist

Research Stage Assessment

For items 1-4, give a score from the scale below that defines the role of AI in each part of the scientific process. The scores are as follows:

- **[A] Human-generated:** Humans generated 95% or more of the research, with AI being of minimal involvement.
- **[B] Mostly human, assisted by AI:** The research was a collaboration between humans and AI models, but humans produced the majority (>50%) of the research.
- **[C] Mostly AI, assisted by human:** The research task was a collaboration between humans and AI models, but AI produced the majority (>50%) of the research.
- **[D] AI-generated:** AI performed over 95% of the research. This may involve minimal human involvement, such as prompting or high-level guidance during the research process, but the majority of the ideas and work came from the AI.

For each research stage where AI was involved, i.e. where you selected **[B]**, **[C]**, or **[D]**, please also indicate the approximate level of iteration effort required using the provided iteration macros. Count substantive attempts, prompts, runs, agent trajectories, or course corrections; do not count minor wording edits.

- **[Low]** : approximately 1-10 substantive attempts.
- **[Medium]** : approximately 10-100 substantive attempts, with some failed attempts or course corrections.
- **[High]** : approximately 100+ substantive attempts, with substantial exploration, failures, or trial-and-error.
- **[NA]** : not applicable because AI involvement was **[A]**.
- **[Unclear]** : the authors cannot reasonably estimate.

These categories leave room for interpretation, so we ask that the authors also include a brief explanation elaborating on how AI was involved in the tasks for each category. Please keep your explanation to less than 150 words.

1. **Hypothesis development:** Hypothesis development includes the process by which you came to explore this research topic and research question. This can involve the background research performed by either researchers or by AI. This can also involve whether the idea was proposed by researchers or by AI.

Answer: **[B]**

Iteration Effort: **[Medium]**

Explanation: The research direction (uncertainty quantification for a structural neural surrogate) was identified through extended human–AI discussion spanning multiple sessions. The human author brought the domain expertise (mechanical engineering, bracket sizing practice) and the strategic framing (deployment reliability rather than surrogate accuracy as the contribution). The AI partner contributed literature grounding, landscape analysis of UQ methods, and helped narrow from a broad “physical AI” theme to the specific L-bracket UQ formulation. Two earlier project directions were abandoned after substantive investigation before the present scope was settled; the final choice of a GNN + Deep Ensemble + CQR stack emerged from iterated human–AI deliberation rather than a single prompt.

2. **Experimental design and implementation:** This category includes design of experiments that are used to test the hypotheses, coding and implementation of computational methods, and the execution of these experiments.

Answer: **[C]**

Iteration Effort: **[High]**

Explanation: An AI coding agent implemented the full pipeline: the FEniCSx FEA driver, the Latin-Hypercube parametric sweep, the MeshGraphNets-style GNN, the Deep Ensemble

training loop, the split-conformal calibration, the pre-registered OOD sweep, and the deferral decision rule. The human author made the architectural and methodological decisions (GNN vs. CNN, 5-member ensemble size, CQR over MC Dropout, parameter ranges, material selection, single load case, 2D plane-stress reduction) and directed the agent to execute them. High iteration count: four failed attempts at a remote-compute sweep (install hang, captured-stdout silence, browser-lag firehose, recursive spawn deadlock) preceded the pivot to local WSL2 execution; the final local pipeline was reached after dozens of debugging cycles.

3. **Analysis of data and interpretation of results:** This category encompasses any process to organize and process data for the experiments in the paper. It also includes interpretations of the results of the study.

Answer: [B]

Iteration Effort: [Medium]

Explanation: The AI agent generated all analysis scripts, figures, tables, and summary statistics (accuracy metrics, coverage sweeps, conditional coverage tables, OOD breakdown, deferral ROC). The human author drove interpretation: what the sample-level Pearson 0.944 implies for deployment trust, how the peak-stress MAPE blow-up on OOD corner samples should be framed, which extrapolation axes should be called “safe” in the Discussion. Follow-up analyses (per-condition breakdown, conditional coverage by design-parameter quartile) were directed by the human after critically reviewing first-pass results.

4. **Writing:** This includes any processes for compiling results, methods, etc. into the final paper form. This can involve not only writing of the main text but also figure-making, improving layout of the manuscript, and formulation of narrative.

Answer: [C]

Iteration Effort: [Medium]

Explanation: The AI agent drafted all sections of the paper progressively as each phase (surrogate, CQR, OOD) was completed. The human author reviewed every section, directed revisions, and made editorial decisions on terminology (“reliability layer” framing, “defer-to-FEA” naming), on what to emphasize in the abstract and introduction, and on the structure of the Limitations discussion. The narrative — that the contribution is the calibrated reliability layer rather than the surrogate itself — was shaped through iterated human–AI discussion during Phase 1 and held throughout.

AI System Documentation

For items 5-6, provide a free-text description.

5. **AI systems used:** Describe all AI systems used in this research without naming specific products or models. Use system-level descriptors only (e.g., “a frontier large language model”, “a protein structure prediction system”, “a custom multi-agent framework built on open-source models”). Include relevant details such as system type, scale, and capabilities. **Specific model and product names should only be included in the camera-ready version.**

Description: Three AI systems were used. (1) A frontier large language model accessed through a web-based chat interface was used for research planning, literature review, methodology design discussions, and iterative scoping of the project. (2) A command-line coding agent built on the same class of frontier large language model was used for all computational implementation: FEA pipeline development, parametric data generation orchestration, GNN architecture implementation, ensemble training, CQR calibration, OOD evaluation, figure and table generation, and paper drafting. The coding agent operates with filesystem, shell, and git access and was run interactively under human direction. (3) A persistent knowledge-graph literature system was used during some sessions to ground citations and methodological choices against a curated corpus of prior work.

6. **Observed AI Limitations:** What specific limitations and failure modes have you observed when using AI as a partner or lead author?

Description: Several limitations were observed and are documented in the project’s agent log. (i) *Compute-time mis-estimation*. The coding agent estimated per-sample FEA cost at 15–30 seconds, motivating a remote-compute plan; the measured local cost was 0.39 seconds

343 per sample, a $40\text{--}75\times$ over-estimate. Acting on the wrong estimate cost roughly two days
344 of detour through remote-platform infrastructure before the human directed a pivot to local
345 execution. (ii) *Repeated failures on unfamiliar infrastructure*. The agent attempted a remote
346 Kaggle-hosted sweep four times in succession, each failing for a different reason (a PyVista
347 install hang, a `capture_output` stdout silence bug, a browser-lag log firehose, and a
348 recursive multiprocessing spawn deadlock). The agent did not reliably diagnose the class of
349 failure on its own; the human-in-the-loop two-attempt rule was what forced the architectural
350 pivot. (iii) *Tooling instability*. The knowledge-graph literature tool was unavailable in one
351 session due to a configuration issue that the agent could not self-diagnose. (iv) *Numerical*
352 *inconsistency across turns*. The agent occasionally reported slightly different summary
353 statistics across messages in a single session, requiring the human to re-run scripts and pin
354 the canonical numbers rather than trust the chat transcript. (v) *Scope creep on writing*. The
355 agent’s first drafts tended to over-claim on generality and required active trimming to match
356 the actual (single geometry, single material, 2D, linear) scope.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and Section 1 state the contribution as the reliability layer (Deep Ensembles + CQR + deferral rule) on top of a MeshGraphNets-style surrogate, explicitly scope the study to a 2D plane-stress L-bracket on a single material, and preview the OOD-extrapolation results. Headline numbers in the abstract match Sections 4.

Guidelines:

- The answer NA means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
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- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: The Discussion section enumerates six limitations: marginal (not conditional) coverage, calibration–selection set reuse, cell-wise exchangeability assumption, absence of an aleatoric head, idealized physics, and bounded OOD protocol scope. Computational scalability is discussed in Section 5: the UQ pipeline adds only constant multiplicative overhead at the current scale (3 parameters, $\sim 3,100$ nodes), but empirical validation on higher-dimensional parametric families and larger meshes has not been performed and remains a limitation of the present study.

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- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
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3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The only theoretical claim invoked is the Romano 2019 Theorem 1 coverage guarantee for CQR; Section 3.3 states the exchangeability assumption explicitly, cites the source theorem, and does not reprove it. Section 3.4 notes that the assumption fails for the OOD set and therefore the guarantee expires there.

Guidelines:

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- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 and the appendices document the parameter ranges, material constants, boundary and loading conditions, mesh refinement, GNN architecture, training schedule, ensemble seeds, calibration set, CQR score definition, and OOD extrapolation protocol in sufficient detail to re-implement the pipeline; data splits and hyperparameters are specified numerically.

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- If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important.
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5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The full code repository will be released publicly at the camera-ready stage. For anonymized submission, the repository structure and all random seeds are documented in the paper to enable independent reimplementations. The dataset regeneration script reruns the LHS sweep end-to-end in approximately 50 minutes on the reference hardware.

Guidelines:

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6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyper-parameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Section 3.2 specifies optimizer (Adam), initial learning rate (5×10^{-4} cosine-annealed to 10^{-5}), loss (Huber, $\delta = 1$ MPa), batch size (8), budget (60 epochs, patience 12), architecture ($H=128$, $L=5$). Table 1 specifies the 4000/500/500/250 train/val/test/OOD partition and the single-use test-set discipline. Computational scalability of the UQ pipeline is discussed in Section 5 (Discussion).

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7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The 5-member Deep Ensemble is itself the source of variance across seeds, with per-seed breakdown in Appendix B. Coverage is reported over 500 test samples (node-pooled across $\sim 1.56 \times 10^6$ nodes), with conditional coverage per quartile. Phase-3 metrics are computed on a 250-sample pre-registered OOD set with per-condition breakdown.

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8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Section 3.2 states that all training is performed on a single NVIDIA RTX 4060 Laptop GPU (8 GB VRAM) in a 24 GB system-RAM Windows environment. FEA data generation runs on the same laptop under WSL2 and the full 5000-sample sweep completes in approximately 50 minutes. Each ensemble member trains in approximately 86 minutes on the stated hardware; the full 5-member ensemble completes in under 8 hours.

Guidelines:

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533 Justification: This is a methodology paper on calibrated uncertainty for a canonical 2D
534 structural-mechanics surrogate; it introduces no new capability in a socially sensitive domain.
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542 (e.g., disinformation, generating fake profiles, surveillance), fairness considerations,
543 privacy considerations, and security considerations.
- 544 • If there are negative societal impacts, the authors could also discuss possible mitigation
545 strategies.

A FEA pipeline details

Geometry and boundary conditions. Hole 1 is centered on the vertical flange at $y = 40$ mm; its bolt-reaction is deliberately omitted (traction-free rim) following standard screening-level sizing practice, which concentrates modelling effort on the geometric concentrators rather than on the fastener detail. The outside corner is kept sharp: being on the compressive outer bend, it is not stress-concentrating. The shelf load acts as a constant downward traction on $W + R \leq x \leq B$, $y = W$; all remaining boundary segments are traction-free. The geometric-feasibility constraint $W + R \leq 34$ mm (from Hole-1 clearance) binds the joint upper corner of the sweep box.

Discretization. FEniCSx with Gmsh-generated triangular meshes. Displacements on Lagrange order-2 elements; stresses recovered by differentiation onto a scalar Lagrange-2 space and combined pointwise as $\sigma_{\text{vM}} = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\sigma_{xy}^2}$ (plane-stress von Mises). Element size drops from $h_{\text{fine}} = 0.2$ mm near the three concentrators to $h_{\text{coarse}} = 2.0$ mm in the far field over an 8 mm transition.

Mesh convergence. At the nominal sample $(R, p, W) = (6.5, 57.0, 19.0)$, four refinement levels were swept (Table 4). The finest level (adopted) carries $\sim 11,000$ scalar DOFs and peak stress changed $+0.024\%$ vs. the previous level, well below the 1% target.

Table 4: Mesh convergence at the nominal sample, unit load ($w = 1$ MPa). Level 3 is adopted as the production refinement.

Level	h_{coarse}	h_{fine}	DOFs	Peak σ_{vM} [MPa]	Δ vs. prev.
0	4.0	1.5	1 398	45.146	–
1	3.0	0.8	2 582	45.587	+0.97%
2	2.5	0.4	5 139	45.632	+0.10%
3	2.0	0.2	11 234	45.643	+0.024%

Analytical cross-check. An un-notched L (same outline, no holes, sharp inside corner) meshed at production refinement and solved at unit load was compared to the Euler–Bernoulli cantilever prediction $\sigma_b(x) = 3w(B - x)^2/W^2$. At $x = 34$ mm, $W = 19$ mm, $w = 1$ MPa the analytical value is 17.58 MPa versus 17.91 MPa FEA, a $+1.8\%$ deviation within expected mesh-refinement bias near the sharp corner.

B Per-seed ensemble metrics

Table 5: Per-seed and ensemble-mean accuracy on the ID test set.

Model	Per-node MAPE [%]	Peak-stress MAPE [%]	p_{50} [MPa]	p_{90} [MPa]	p_{99} [MPa]	max error [MPa]
Seed 0	2.45	0.43	0.044	0.184	0.59	2.57
Seed 101	2.62	0.75	0.049	0.178	0.41	2.14
Seed 202	4.13	0.65	0.066	0.288	0.80	3.71
Seed 303	3.00	1.66	0.054	0.230	0.82	3.27
Seed 404	3.35	0.50	0.062	0.229	0.55	2.07
Ensemble mean	1.81	0.42	0.032	0.124	0.32	2.13

C Full CQR coverage sweep and conditional coverage

Table 6 extends the three-level coverage summary of Table 2 to the full four nominal levels swept ($\alpha \in \{0.20, 0.15, 0.10, 0.05\}$). Table 7 slices empirical coverage at nominal 90% by quartile of each design parameter, and Figures 7–9 visualize per-sample CQR interval widths, example miscoverage masks, and the width–vs.–error correspondence that motivates the deferral rule in Section 3.5.

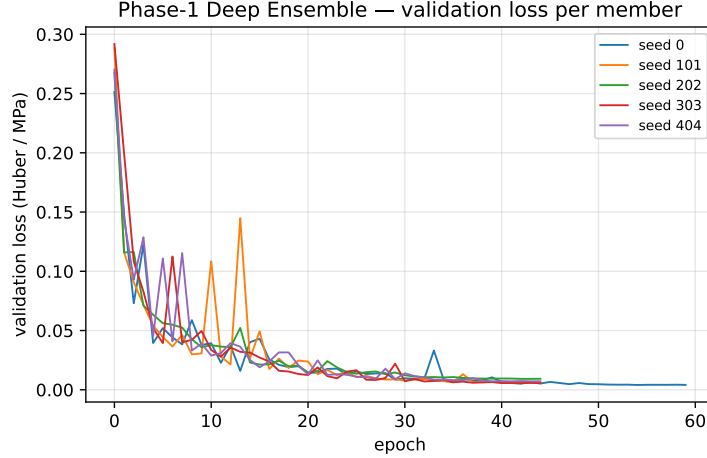


Figure 6: Validation loss per epoch for the five Deep-Ensemble members (seeds 0, 101, 202, 303, 404). Huber loss on the un-normalized stress target in MPa; cosine schedule with early-stopping patience 12.

Table 6: Full four-level CQR coverage sweep (extends Table 2).

Nominal [%]	Uncalibrated (Gaussian)		CQR-calibrated		$\hat{Q}$ [MPa]
	Coverage	Width [MPa]	Coverage	Width [MPa]	
80	86.41	0.25	82.72	0.24	−0.005
85	89.44	0.28	87.03	0.27	−0.004
90	92.32	0.32	91.28	0.31	−0.003
95	95.19	0.38	95.53	0.38	0.001

573 D OOD per-condition breakdown

574 Table 8 reports accuracy, CQR coverage, mean interval width, ensemble std, and deferral-rule
575 detection rate for each of the seven pre-registered OOD conditions (six single-parameter directions
576 plus the corner condition where two or more parameters are simultaneously outside the training
577 box). Figures 10–12 visualize the ID-vs-OOD comparison, the per-direction breakdown, and three
578 representative OOD predictions (best, median, worst by mean absolute error).

579 E Deferral rule operating characteristic

580 Table 9 reports the ID false-alarm and OOD detection rates at four candidate thresholds (ID-std
581 quantiles 80%, 90%, 95%, 99%); the deployment operating point is the 95th percentile, which by
582 construction yields 5% ID false-alarm. Figure 13 shows the corresponding ROC of per-sample mean
583 ensemble std as an OOD detector, together with the density of ensemble-std values for ID vs. OOD
584 samples.

585 F Vanilla split-conformal comparison

586 As a minimal-baseline reference for CQR we evaluate vanilla split-conformal prediction: taking
587 $E_i = |\bar{\sigma} - \sigma^*|$ on the validation set as the nonconformity score, calibrating $Q_{\text{vanilla}}(\alpha)$ via the
588 standard finite-sample corrected quantile, and constructing constant-width symmetric intervals $[\bar{\sigma} -$
589 $Q_{\text{vanilla}}, \bar{\sigma} + Q_{\text{vanilla}}]$ on the test set. This interval cannot adapt to local heteroscedasticity by
590 construction (Table 10).

591 Vanilla split-conformal produces narrower intervals but under-covers the nominal level: the $\alpha = 0.10$
592 interval achieves 81.1% empirical coverage instead of 90%. The Romano Theorem 1 guarantee still

Table 7: Conditional coverage at nominal 90%, sliced by quartile of each design parameter. Coverage stays within ± 3 pp of nominal across all 12 quartiles.

Slice	Range	n samples	Coverage [%]	Mean width [MPa]
R Q1	[6.51, 7.43]	125	91.89	0.26
R Q2	[7.43, 8.42]	125	92.16	0.33
R Q3	[8.42, 9.26]	125	91.78	0.31
R Q4	[9.26, 10.00]	125	89.33	0.34
p Q1	[57.02, 61.83]	125	91.44	0.20
p Q2	[61.83, 66.19]	125	91.91	0.29
p Q3	[66.19, 69.08]	125	91.01	0.33
p Q4	[69.08, 72.00]	125	90.76	0.43
W Q1	[19.01, 20.92]	125	90.80	0.20
W Q2	[20.92, 22.03]	125	92.67	0.29
W Q3	[22.03, 23.00]	125	91.71	0.34
W Q4	[23.00, 24.00]	125	89.95	0.41

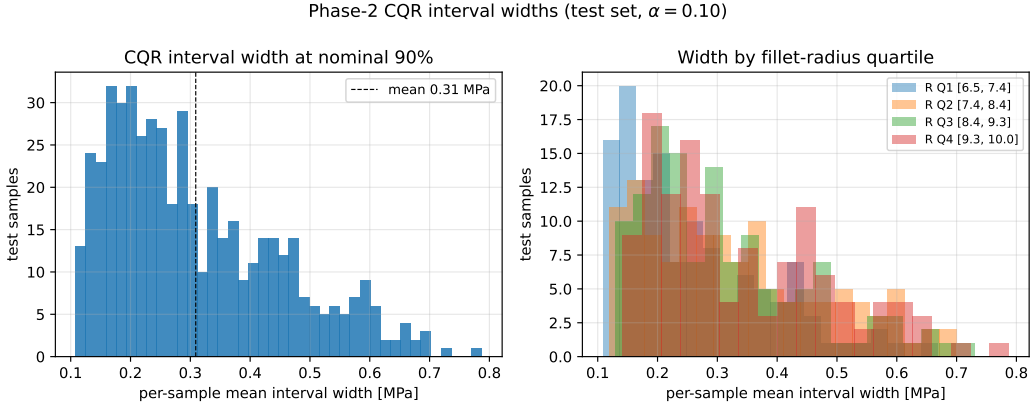


Figure 7: Distribution of per-sample mean CQR interval widths on the test set at nominal 90%. *Left:* pooled over all 500 test samples. *Right:* stratified by fillet-radius quartile.

holds at the point level of exchangeable units, but the cell-wise pooling used here treats mesh nodes within a graph as exchangeable when they are in fact correlated within each sample. CQR’s adaptive width — wider on high-uncertainty regions and samples — absorbs this within-sample correlation and recovers nominal coverage. This is exactly the deployment-relevant property CQR adds beyond the simplest conformal baseline: adaptive intervals that remain valid when the exchangeability assumption is only approximately met.

G Ensemble-size ablation

We re-use the frozen 5-member ensemble to evaluate the effect of ensemble size $M \in \{2, 3, 4, 5\}$ on accuracy and CQR coverage at $\alpha = 0.10$. For each M , we take the first M seeds from $\{0, 101, 202, 303, 404\}$, recompute the ensemble mean and std, re-run the CQR calibration on the 500-sample validation set, and evaluate on the test set (Table 11). No retraining is performed.

Both accuracy (per-node MAPE) and calibration quality (CQR coverage at 90%) improve monotonically from $M = 3$ onward. $M = 5$ delivers the best per-node MAPE and the best CQR coverage, confirming the Lakshminarayanan [5] recommendation of 5-member ensembles as a robust practical choice. The mean ensemble std grows with M as expected (small- M ensembles under-estimate variance), which explains the widening CQR interval with M ; the net effect is better-calibrated coverage rather than over-wide intervals.

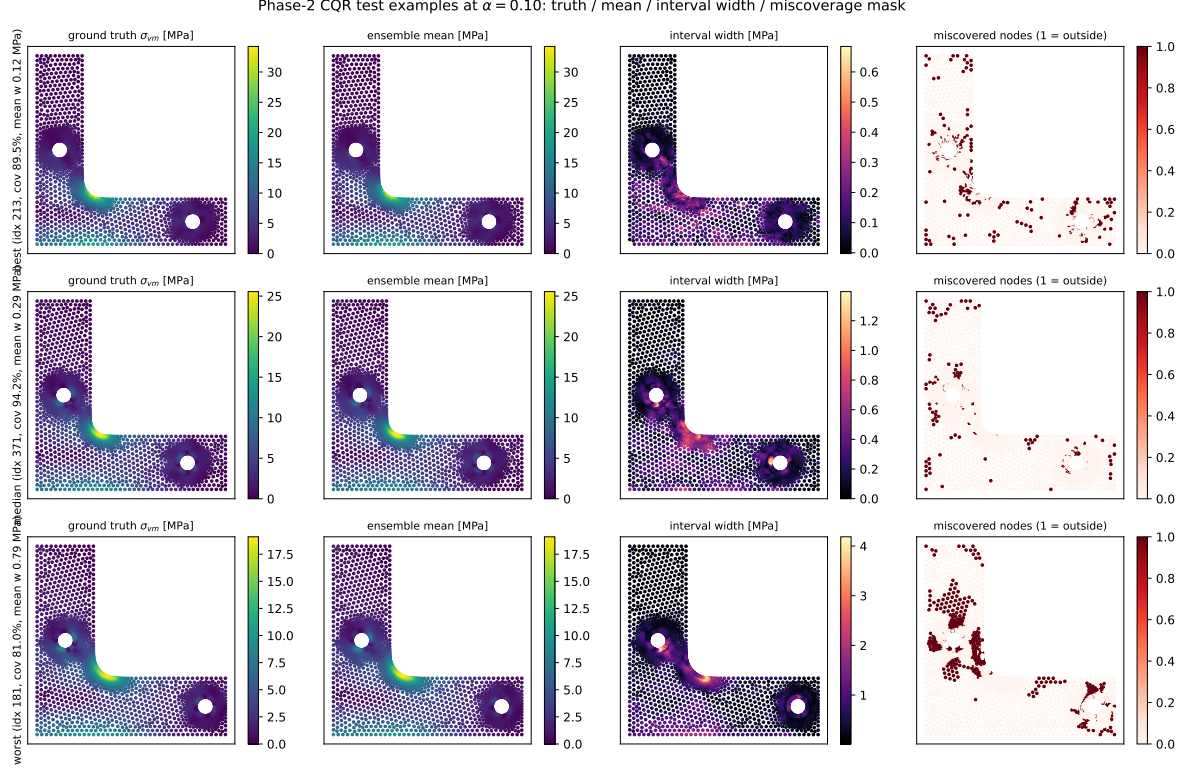


Figure 8: CQR test-set examples at $\alpha = 0.10$ (best, median, worst per-sample error). Columns: ground-truth stress; ensemble-mean prediction; per-node CQR interval width; miscoverage mask (red = true stress outside interval).

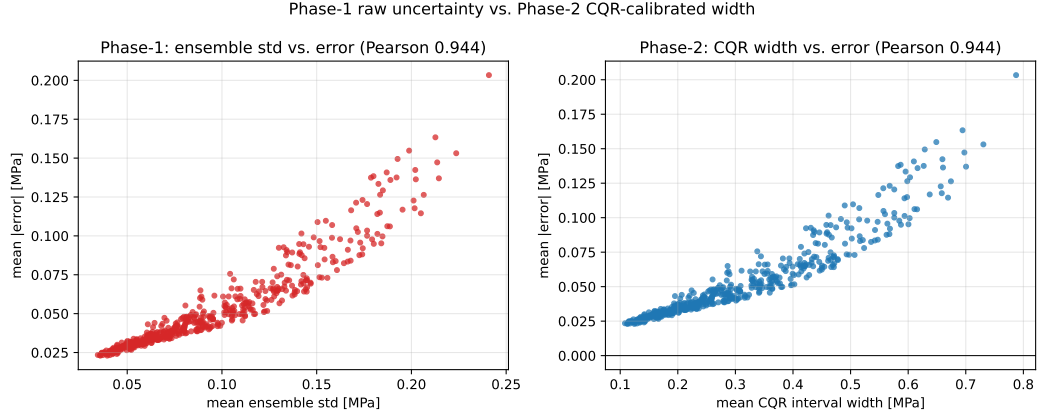


Figure 9: Phase-1 raw ensemble-std uncertainty (left) vs. Phase-2 CQR-calibrated interval width (right), plotted against per-sample mean absolute error on the test set.

Table 8: Per-condition OOD breakdown at nominal 90% coverage.

Condition	n	Per-node MAPE [%]	Coverage [%]	Width [MPa]	Ens. std [MPa]	Detection [%]
R -low	25	1.08	88.10	0.255	0.079	0.0
R -high	25	1.29	89.21	0.249	0.077	0.0
p -low	25	2.72	84.27	0.663	0.203	44.0
p -high	25	2.72	81.36	0.242	0.075	4.0
W -low	25	2.63	78.17	0.658	0.201	44.0
W -high	25	2.35	88.77	0.369	0.114	16.0
Corner (≥ 2 OOD)	100	4.56	81.46	1.085	0.332	58.0

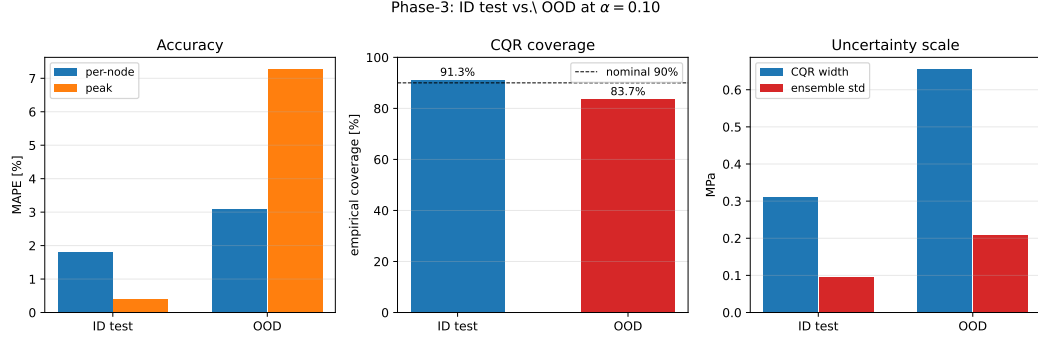


Figure 10: ID test vs. OOD at nominal 90% coverage: per-node and peak-stress MAPE, empirical coverage, and interval width / ensemble std.

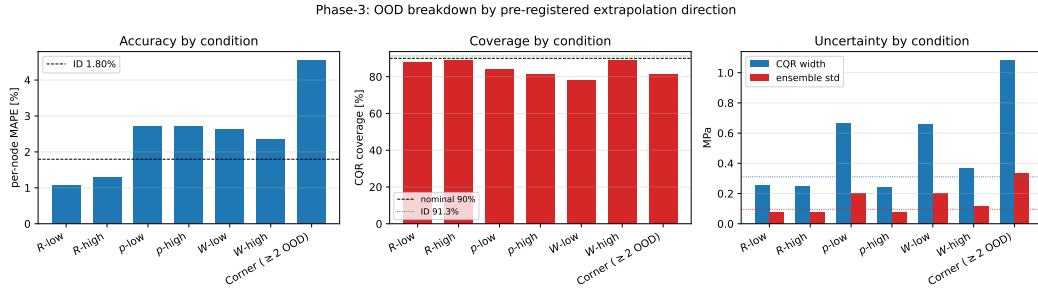


Figure 11: OOD metrics broken down by pre-registered extrapolation direction. *R*-direction extrapolation is handled well; *p*-low, *W*-low, and corner OOD drive the aggregate degradation.

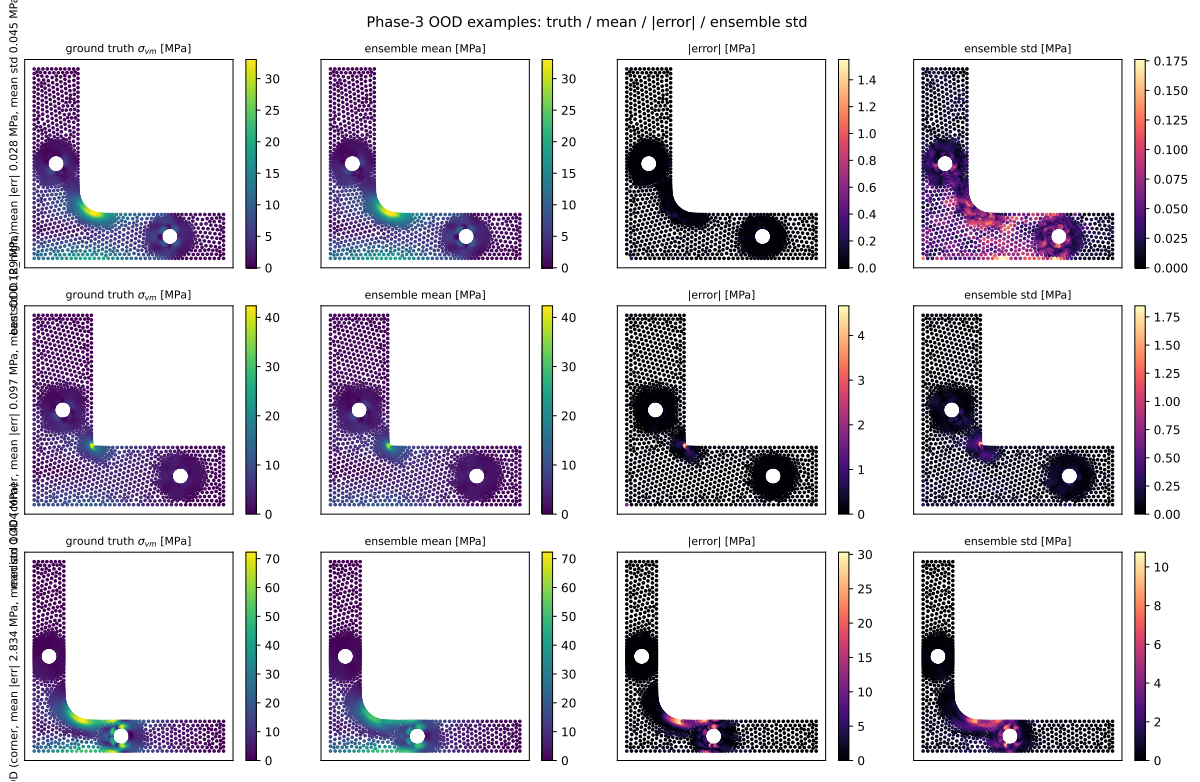


Figure 12: OOD example stress fields (best, median, worst by mean absolute error). Error and std maps are visibly broader than ID and migrate off the fillet onto the full bulk field in the worst sample.

Table 9: Deferral operating points at four candidate ID-std quantile thresholds. The ID false-alarm row is $1 - q$ by construction.

Threshold (ID quantile)	ID false-alarm [%]	OOD detection [%]
80% ($T = 0.134$ MPa)	20.0	45.2
90% ($T = 0.160$ MPa)	10.0	37.2
95% ($T = 0.182$ MPa)	5.0	34.0
99% ($T = 0.207$ MPa)	1.0	27.6

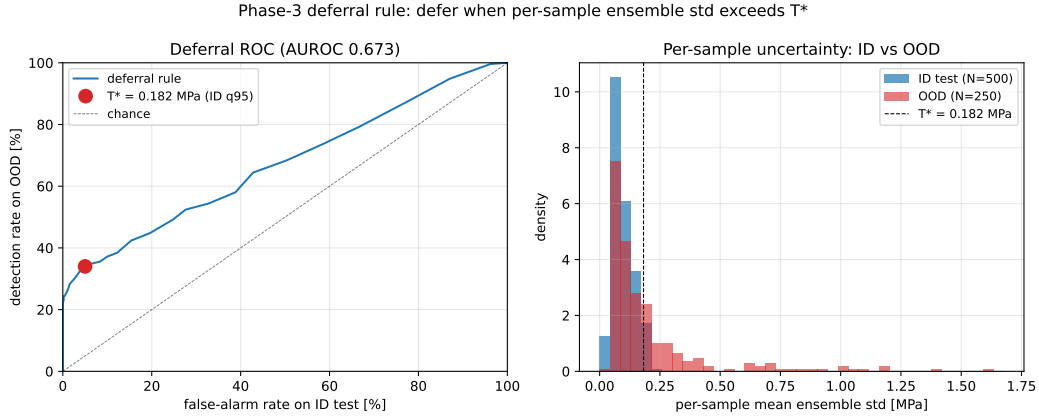


Figure 13: Defer-to-FEA decision rule. *Left*: ROC of the per-sample mean ensemble std as an OOD detector; the marker shows the deployment operating point at $T^* = \text{ID-std } 95^{\text{th}}$ percentile. *Right*: density of per-sample ensemble std for ID vs. OOD.

Table 10: Vanilla split-conformal (constant-width) vs. CQR (adaptive-width) on the ID test set, cell-wise pooled. $2\hat{Q}$ and width are reported in MPa.

Nominal [%]	Vanilla split-conformal			CQR	
	Coverage [%]	Q_{vanilla} [MPa]	$2\hat{Q}$ width [MPa]	Coverage [%]	Mean width [MPa]
80	68.66	0.056	0.111	82.72	0.236
90	81.07	0.084	0.167	91.28	0.311
95	89.02	0.118	0.236	95.53	0.379

Table 11: Ensemble-size ablation on the ID test set at nominal 90% coverage. $\hat{Q}$ and width are reported in MPa.

M	Per-node MAPE [%]	Peak-stress MAPE [%]	CQR coverage [%]	Mean width [MPa]	Mean ens. std [MPa]
2	1.98	0.32	86.45	0.252	0.050
3	2.10	0.32	88.35	0.292	0.079
4	1.89	0.52	90.40	0.308	0.091
5	1.81	0.42	91.28	0.311	0.096